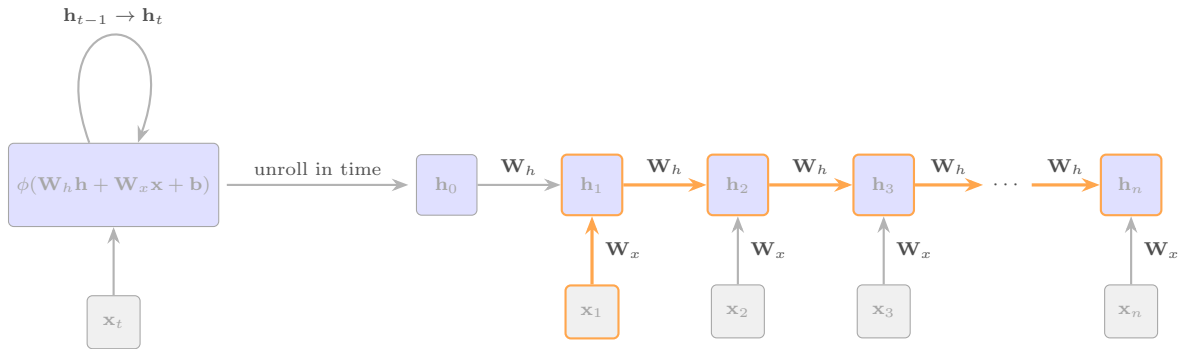




The same tied weights ($\mathbf{W}_h, \mathbf{W}_x, \mathbf{b}$) act at every step, so unrolling the self-loop across n tokens turns one cell into a depth- n feed-forward graph. The only route from $\mathbf{x}_1$ to $\mathbf{h}_n$ (highlighted) is the state chain: it must cross every intermediate $\mathbf{h}_t$ — exactly what makes long-range dependencies costly to learn.